SNEHAL A. KANASE

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Bioinformatician | Cancer Genomics | NGS Data Analysis | Variant Interpretation | Python | R | Linux

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PROFESSIONAL SUMMARY

Bioinformatician with 2 years of combined experience in NGS data analysis, cancer genomics, clinical variant interpretation, and reproducible bioinformatics workflow development. Hands-on experience with WES/WGS, RNA-seq and miRNA-seq data analysis, variant calling, annotation and interpretation using Python, R, Linux, GATK, Samtools and Nextflow. Experienced in developing reproducible workflows using Nextflow and Docker, performing statistical and biological data analysis, and supporting genomics research and manuscript preparation. Strong interest in cancer genomics, clinical genomics, and multi-omics research.

PROFESSIONAL EXPERIENCE

❖ **Freelance Bioinformatics Mentor | *Shivaji University, Kolhapur (Part-time)***

March 2026 – Present (6 Months)

- Mentored students enrolled in specialized Bioinformatics courses on NGS data analysis pipelines, clinical variant interpretation, and computational biology workflows.
- Guided mentees on software engineering best practices, including version control with Git/GitHub, Linux command-line tools, and scripting in Python and R.
- Conducted regular code reviews, debugged custom execution scripts, and resolved pipeline deployment errors for high-throughput sequencing datasets (WES/WGS, RNA-seq).

❖ **Bioinformatician | KIMS Foundation & Research Center, Hyderabad**

Sep 2024 – Feb 2026 (Contract)

- Performed NGS data analysis including quality control, alignment, variant calling, annotation, and interpretation.
- Processed genomic sequencing data using GATK, Samtools, FastQC, Trimmomatic, and BEDTools.
- Conducted variant interpretation, mutation impact assessment, and oncogenomic analyses.
- Developed and optimized reproducible NGS workflows using Nextflow.
- Analyzed RNA-seq and miRNA-seq datasets for gene expression studies.
- Performed Sanger sequencing analysis and variant validation.
- Supported research projects through data analysis, literature review, and manuscript preparation.

RESEARCH & PUBLICATIONS

Manuscript Under Review

PIM1 Kinase as a Prognostic, Expression-Driven Biomarker in Adult Diffuse Gliomas: First Indian Cohort Analysis with Global Validation and Molecular Dynamics Insights

Contributions: NGS and miRNA-seq data analysis, biomarker discovery, expression-based prognostic analysis, molecular dynamics simulations, cancer genomics interpretation, and manuscript preparation.

TECHNICAL SKILLS

Programming & Versioning: Python, R, Bash, Shell Scripting, SQL, Linux/Unix

Pipeline & Automation: Nextflow, Docker, HPC/Cloud

NGS & Genomic Data Analysis: WES/WGS, RNA-seq, Somatic & Germline Variant Calling, Genomic Data Processing, BWA, GATK, Samtools, FastQC, Trimmomatic, STAR/HISAT2, featureCounts, DESeq2, BEDTools

Variant Interpretation & Annotation: ACMG/AMP Classification, Variant Filtering, Annotation & Interpretation, Clinical Variant Interpretation, Mutation Impact Assessment, SIFT, PolyPhen, PROVEAN, Align-GVGD, MetaSNP, ClinVar, gnomAD, Ensembl, Basic CNV & Structural Variant Analysis, Cancer Genomics

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Statistical Data Analysis

Structural Bioinformatics: AlphaFold, I-TASSER, GROMACS, PyMOL

PROJECT

Decoding Late-Onset Type 2 Diabetes through Differential Gene Expression Analysis

Jan 2024 – Mar 2024

- Performed RNA-seq differential gene expression analysis.
- Identified disease-associated genes and biological pathways.
- Applied statistical and visualization techniques for biological interpretation.

EDUCATION

PG Diploma in Bioinformatics - Bversity, Chennai | Aug 2023 – Feb 2024

M.Sc. Industrial Microbiology - Shivaji University, Kolhapur | 2023 | CGPA: 7.9

B.Sc. Microbiology **-** Shivaji University, Kolhapur | 2021 | CGPA: 8.7

CERTIFICATION

Data Science & Python Programming – CodeWithHarry

Key Skills: Python, Pandas, NumPy, Matplotlib, **SQL, Git, Machine Learning Fundamentals**